Week 1 Module 1 Assignment 2

Moffat Bay: TDD Assignment

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Bellevue University

CSD460-H323 Capstone in Software Development (2265-DD)

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Week 1 Module 1 Assignment 2

Moffat Bay: TDD Assignment

GitHub Repository Link:

https://github.com/Joelatkinson95/Moffat-Bay

Group Team Members:

Joel Atkinson, Zachary Baker, Kyle Klausen, Juan Macias Vasquez, Brittaney Perry-Morgan

Leader Role: Joel Atkinson

Moffat Bay Lodge

Group Roles for this Assignment

Purpose	Zachary Baker
Terminology	Brittaney Perry-Morgan
User Personas	Joel Atkinson
User Stories	Juan Macias Vasquez
Work Estimations (ToDo List)	Kyle Klausen

TDD Structure:

- Introduction
  - Purpose
  - Terminology
  - User Personas
  - User Stories
  - Work Estimations (ToDo List)
- Design (Not done yet)
  - 2.1 Prototypes
  - 2.2 ERD
- QA Testing (Not done yet)
  - 3.1 QA Test Plan

Introduction

Purpose

The San Juan Islands First Nations Development Committee has approved construction of a resort at Moffat Bay on Joviedsa Island, and construction is nearly complete. There are concerns that their reach and customer base may be hindered by a lack of online presence. As a result, a website showcasing the Lodge's services has become a necessity. The San Juan Islands First Nations Development Committee requires a website that allows lodge reservation management, user log-in, a database for reservation information, and a dynamically updated Reservation Lookup page.

These services, when tied together, create the foundation for reservations to proceed smoothly while granting users efficient access to listings. Users being able to view the reservation openings reduces the chances of accidental double-bookings while also streamlining the reservation process. Establishing the service in this manner enables the customer to quickly and efficiently navigate the reservation process.

Terminology

Guest — A user who has registered an account on the Moffat Bay platform and may search for and book lodge accommodations.

Reservation — A confirmed lodge room booking associated with a guest account, check-in/check-out dates, room type, and a unique confirmation number.

Lodge — The hotel facility at Moffat Bay Resort on Joviedsa Island, whose rooms are bookable through the web application.

Reservation Lookup — A dynamically updated page allowing guests or staff to search for and retrieve existing reservation records.

User Registration — The process by which a new guest creates an account on the platform, providing credentials stored securely in the database.

Session — A server-side object that maintains authenticated user state across HTTP requests, managed via Jakarta EE session handling.

Servlet — A Java class that processes HTTP requests and responses, functioning as the controller layer of the application architecture.

JSP (JavaServer Pages) — The view layer technology used to render dynamic HTML content returned to the client browser.

JavaBean — A Java class following standard naming and encapsulation conventions, used to pass data between the servlet and JSP layers.

JDBC (Java Database Connectivity) — The API used to execute parameterized queries against the MySQL database.

ERD (Entity Relationship Diagram) — A diagram representing the database schema, including tables, attributes, and the relationships between them.

TDD (Technical Design Document) — The living project document capturing requirements, personas, user stories, design artifacts, and QA planning for the Moffat Bay application.

Kanban Board — The GitHub Projects board used by the team to track task progress across Check In, To Do, In Progress, and Done columns.

User Personas

Moffat Bay Resort & Marina

User Persona

A comprehensive view of Moffat Bay Lodge & Marina's target users, their needs, behaviors, and interactions with the reservation system

Core Profile

Name: Henderson Family

Age and gender: 42, 40, 15, 12

Job: Project Manager & School Teacher

Income range: $120,000 - $150,000

Location: Sacramento, California

Tech Skills: Excellent

Background

Short bio:

Family of 4 from Sacramento CA. Annual vacation planners who premium destinations with relaxation and adventure. Detail orientated and research heavy before booking

Features They Will Use:
- Lodge reservation page
- Attractions page

Features They Won't Use:
- Reservation lookup page

Lifestyle and values:
- Family First, experience driven travel
- Value quality and premium service over cost
- Plan trips months in advance
- Will contact to get as much info and details as possible before booking

Audience Insights

Goals:
- Short-term goals:
  - Book King or double queen room
  - Info on airport & transport
- Long-term goals:
  - Memorable family vacation
  - Potenital return anually
- Key drivers and priorities:
  - Luxury amenities
  - Easy booking and FAQs page

Barriers:
- Practical challenges:
  - Unsure how to get to the island
  - Concerned about kids activities
- Emotional frustrations:
  - Anxious about spending large amount without enough info upfront
- Obstacles to success:
  - Lack of clear transportation info
  - Unclear cancellation policy

Moffat Bay Resort & Marina

User Persona

A comprehensive view of Moffat Bay Lodge & Marina's target users, their needs, behaviors, and interactions with the reservation system

Core Profile

Name: Mary Peters

Age and gender: 32 Female

Job: Registered Nurse

Income range: $100,000

Location: San Juan Islands, Washington

Tech Skills: Good

Background

Short bio:

Lifelong Joviedsa Island resident and First Nations community member. Wants to support the resort while ensuring locals are recognized and welcomed

Features They Will Use:
- About Us page
- Lodge Reservation page

Features They Won't Use:
- Contact Us page
- Attractions page

Lifestyle and values:
- Community orientated and culturally proud
- Values sustainability and businesses that give back to community
- Prefers simple and straightforward websites

Audience Insights

Goals:
- Short-term goals:
  - Find locals & nurses discounts
  - Book weekend staycations
- Long-term goals:
  - Encourage locals to use resort
  - Have resort showcase local pride
- Key drivers and priorities:
  - Community recognition
  - Locals pricing & welcoming vibe

Barriers:
- Practical challenges:
  - No clear information on local first nations discounts
- Emotional frustrations:
  - Concerned resort won't feel welcoming to locals or showcase local pride
- Obstacles to success:
  - No community pricing visible
  - No easy way to ask about local accomodations

Moffat Bay
RESORT & MARINA

User Persona

A comprehensive view of Moffat Bay Lodge & Marina's target users, their needs, behaviors, and interactions with the reservation system

Core Profile

Name: Jessica Williams

Age and gender: 26 Female

Job: Front desk employee at lodge

Income range: $40,000 - $45,000

Location: Friday Harbor, Washington

Tech Skills: Excellent

Background

Short bio:

Recent hospitality graduate from Western Washington University. Wokring as front desk coordinator at lodge. Primary user of reservation lookup. Focussed on fast easy check in

Features They Will Use:
- Reservation confirmation page
- Guest registration & login management

Features They Won't Use:
- Landing page
- Attractions & About Us page

Lifestyle and values:
- Hardworking, organized, people orientated
- Takes pride in excellent customer service
- Tech savvy and quick to learn new systems

Audience Insights

Goals:
- Short-term goals:
  - Quickly look up guest reservations by email or ID
- Long-term goals:
  - Build a smooth and welcoming check in experience for guests
- Key drivers and priorities:
  - Fast reliable reservation look up
  - Clear booking details

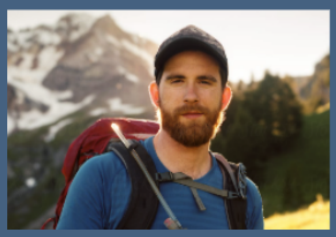
Barriers:
- Practical challenges:
  - Stystem slowdowns during peak check in times
- Emotional frustrations:
  - Embarrassed infront of guests when system fails
- Obstacles to success:
  - Lookup not returning accurate results
  - No fallback search

Moffat Bay
RESORT & MARINA

User Persona

A comprehensive view of Moffat Bay Lodge & Marina's target users, their needs, behaviors, and interactions with the reservation system

Core Profile



Name: Matthew Webb

Age and gender: 40 Male

Job: Freelance graphic designer

Income range: $50,000 - $70,000

Location: Portland, Oregon

Tech Skills: Good

Background

Short bio:

Solo traveler and outdoor enthusiast. Drawn to Moffat Bay for the island's hiking, kayaking, and whale watching. Books trips spontaneously and prefers a fast easy online experience

Features They Will Use:
- Lodge reservation page
- Attractions page

Features They Won't Use:
- Contact us page
- About us page

Lifestyle and values:
- Independent and outdoorsy
- Values unique experiences over luxury
- Comfortable booking and managing everything himself online

Audience Insights

Goals:
- Short-term goals:
  - Book double or queen room for solo weekend trip
  - Learn about the area attractions
- Long-term goals:
  - Add Joviedsa Island to his list of PNW adventures
- Key drivers and priorities:
  - Attractions page
  - Easy reservations & quick booking process

Barriers:
- Practical challenges:
  - Wants to confirm availability before booking
- Emotional frustrations:
  - Concerned resort caters to families and couples not solo travelers.
- Obstacles to success:
  - Limited information on the attractions in the area.

User Stories

Henderson Family (Family Travelers) Persona

Persona insight:

- Wants easy booking
- Needs info about transportation
- Concerned about kids' activities

User Stories:

- US-1 (High Priority)
  "As a family traveler, I want to easily book a king or double queen room, so that my family of four can stay comfortably together."
- US-2 (High Priority)
  "As a family traveler, I want to view transportation options to the island, so that I can plan our trip without confusion or stress."
- US-3 (Medium Priority)
  "As a family traveler, I want to see nearby attractions and kid-friendly activities, so that I can plan fun experiences for my children."

Mary Peters (Local Resident & Registered Nurse) Persona

Persona insight:

- Easy Local Information / Local Pride
- Wants clear information
- Local Pricing

User Stories:

- US-1 (High Priority)
  As a local resident, I want to see any locals or nurse discounts when booking, so that I feel recognized and can take advantage of benefits available to my community.
- US-2 (High Priority)
  As a local resident, I want to book weekend staycations easily, so that I can relax without complicated processes.
- US-3 (Medium Priority)
  As a local resident, I want clear information about resort events that showcase local culture, so that I can support and participate in community-focused activities.
- US-5 (Low Priority)
  As a local resident, I want an easy way to contact resort staff about local accommodations, so that I can ask questions about special local arrangements.

Jessica Williams (Front Desk Employee) Persona

Persona insight:

- Needs a fast and reliable system to quickly check in guests
- Prioritizes accurate reservation details and having an easy search functionality
- Works under pressure during peak times and values efficiency
- Gets frustrated when system delays or errors impact customer experience

User Stories:

- US-1 (High Priority)
  As a front desk employee, I want to quickly search for reservations by guest name or ID, so that I can check in guests efficiently.
- US-2 (High Priority)
  As a front desk employee, I want to view complete reservation details, so that I can confirm bookings accurately.
- US-3 (High Priority)
  As a front desk employee, I want the system to load quickly during peak hours, so that I can avoid delays in guest check-in.
- US-4 (Medium Priority)
  As a front desk employee, I want a backup search option (like phone number), so that I can find reservations if primary search fails (If the wrong name was used).
- US-5 (Low Priority)
  As a front desk employee, I want a simple and easy-to-use interface, so that I can navigate the system without errors.

Matthew Webb (solo traveler & outdoor enthusiast) Persona

Persona insight:

- Prefers a fast and simple booking process for spontaneous trips
- Values clear information about outdoor activities and local attractions
- Likes to manage everything independently online himself without help

User Stories:

- US-1 (Medium Priority)
  As a solo traveler, I want to view nearby outdoor attractions, so that I can plan activities like hiking and kayaking.
- US-2 (Medium Priority)
  As a solo traveler, I want clear pricing and room details, so that I can make quick decisions.
- US-3 (Low Priority)
  As a solo traveler, I want everything to be handled online, so that I don’t need to contact support.

Work Estimations (ToDo List)

Task Category	Task	Estimate Time
User Account	Registration form & validation	4 hours
User Account	Login/Logout functionality	3 hours
User Account	Session management	2 hours
Reservation System	Database schema design	3 hours
Reservation System	Room availability logic	5 hours
Reservation System	Booking functionality	6 hours
Reservation System	Confirmation number system	2 hours
Reservation Lookup	Search feature	4 hours
Reservation Lookup	Display reservation details	3 hours
Frontend	Page development	6 hours
Frontend	Styling & responsiveness	5 hours
Database	JDBC integration	4 hours
Database	CRUD testing	3 hours
Testing	Unit testing	4 hours
Testing	Debugging & optimization	4 hours
	Total:	**58 hours**

Comment: Tasks will be tracked using a Kanban board with columns for To Do, In Progress, and Completed.